

Chapter 5 – System Design

5.1 System Architecture

The CyberCafe Management Hub follows a **three-tier architecture**, which separates the system into three main layers: presentation layer, application layer, and database layer. This structure helps in maintaining the system efficiently and improves scalability and performance.

- **Presentation Layer (Frontend):**

This layer includes the user interface developed using HTML, CSS, and JavaScript. It is responsible for displaying information and taking user input.

- **Application Layer (Backend):**

This layer is developed using PHP. It processes user requests, handles business logic, and communicates between the frontend and the database.

- **Database Layer:**

This layer uses MySQL to store all the data such as faculty details, departments, subjects, attendance, and notices.

This architecture ensures proper separation of concerns and makes the system more organized and easy to maintain.

5.2 Database Design

The database is designed using MySQL and consists of multiple related tables to store and manage data efficiently. Each table is connected through primary and foreign keys to maintain relationships.

Main Tables Used:

- Admins
- Departments
- Subjects
- Faculty
- Attendance
- Leaves
- Timetable
- Notices

Each table stores specific information and is linked to others where necessary. For example, the Faculty table is connected to Departments and Subjects, ensuring structured and organized data storage.



Figure 5.2: DFD Level 0 (Context Diagram)

5.3 Flowchart

The system flowchart represents the overall working process of the application. It starts from the user accessing the system, followed by login authentication, and then navigation through different modules such as faculty management, attendance, timetable, and reports.

The flowchart helps in understanding how data flows within the system and how different modules interact with each other. It ensures that the process is logical and error-free.

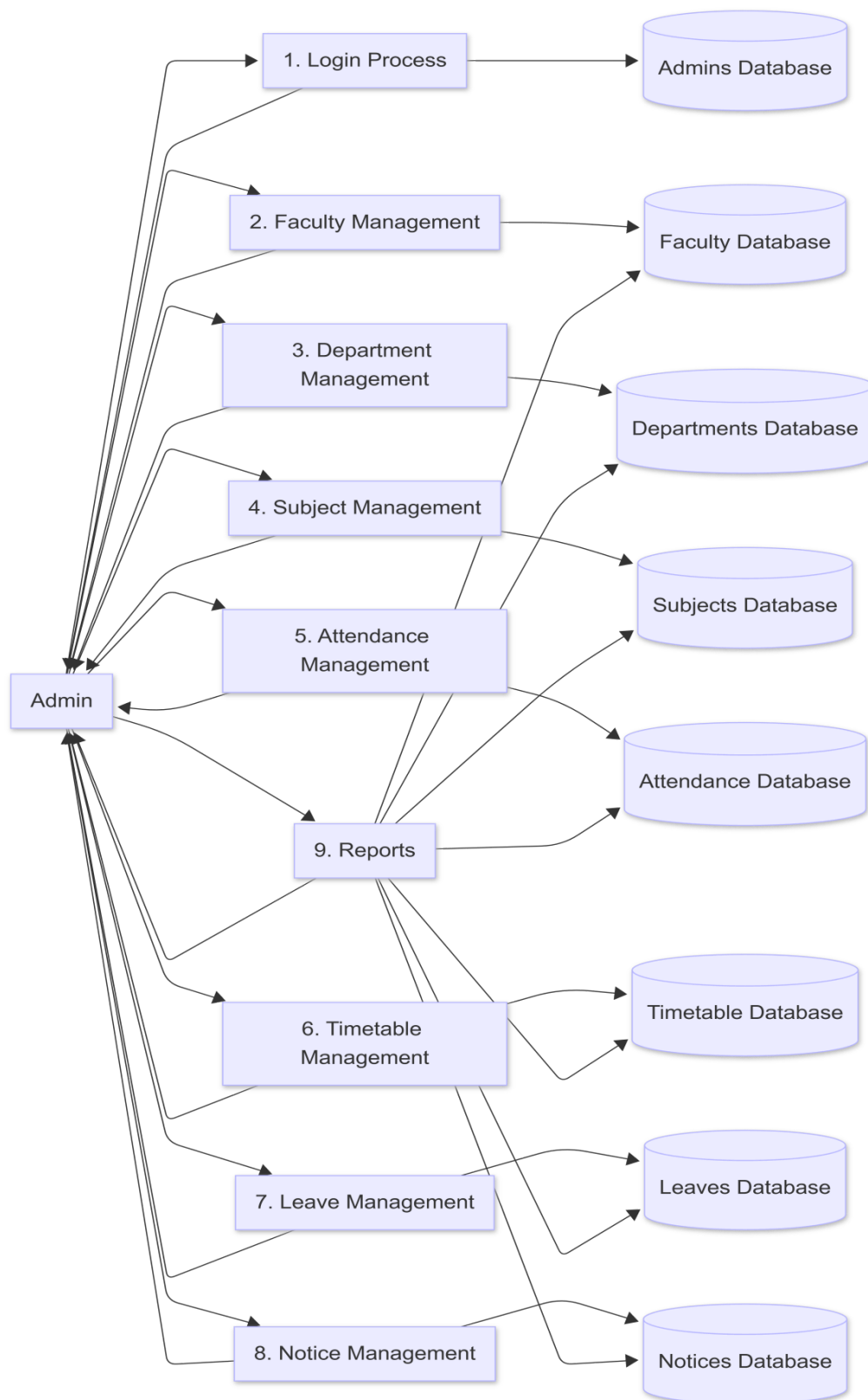


Figure 5.3: DFD Level 1 of Faculty Management Hub System

5.4 ER Diagram (Entity Relationship Diagram)

The ER Diagram represents the relationship between different entities in the database. It shows how tables are connected with each other using keys.

Main Relationships:

- Faculty is linked to Departments
- Faculty is linked to Subjects
- Attendance is linked to Faculty
- Leaves are linked to Faculty
- Timetable is linked to Faculty, Subjects, and Departments

The ER diagram helps in visualizing the database structure and ensures proper data organization without redundancy.

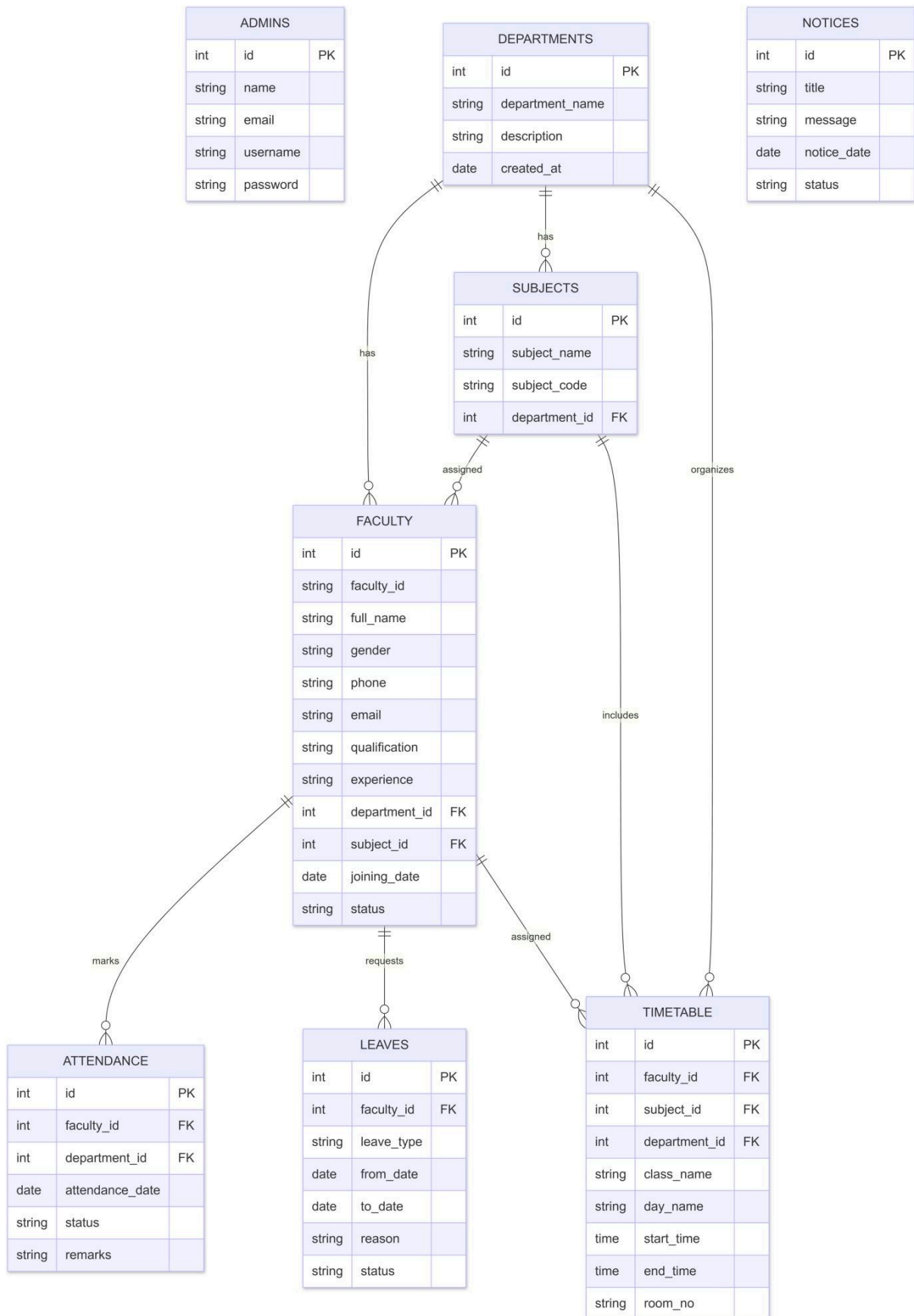


Figure 5.4: ER Diagram of Faculty Management Hub System

5.5 DFD (Data Flow Diagram)

The Data Flow Diagram (DFD) represents how data moves within the system.

It shows the interaction between the user, system processes, and database.

Main Components of DFD:

- **Input:** Admin enters data (faculty details, attendance, etc.)
- **Process:** System processes the data using backend logic
- **Output:** Data is displayed in the form of tables, reports, or dashboards
- **Storage:** Data is stored in the database

The DFD helps in understanding how information flows and how the system processes and stores data efficiently.

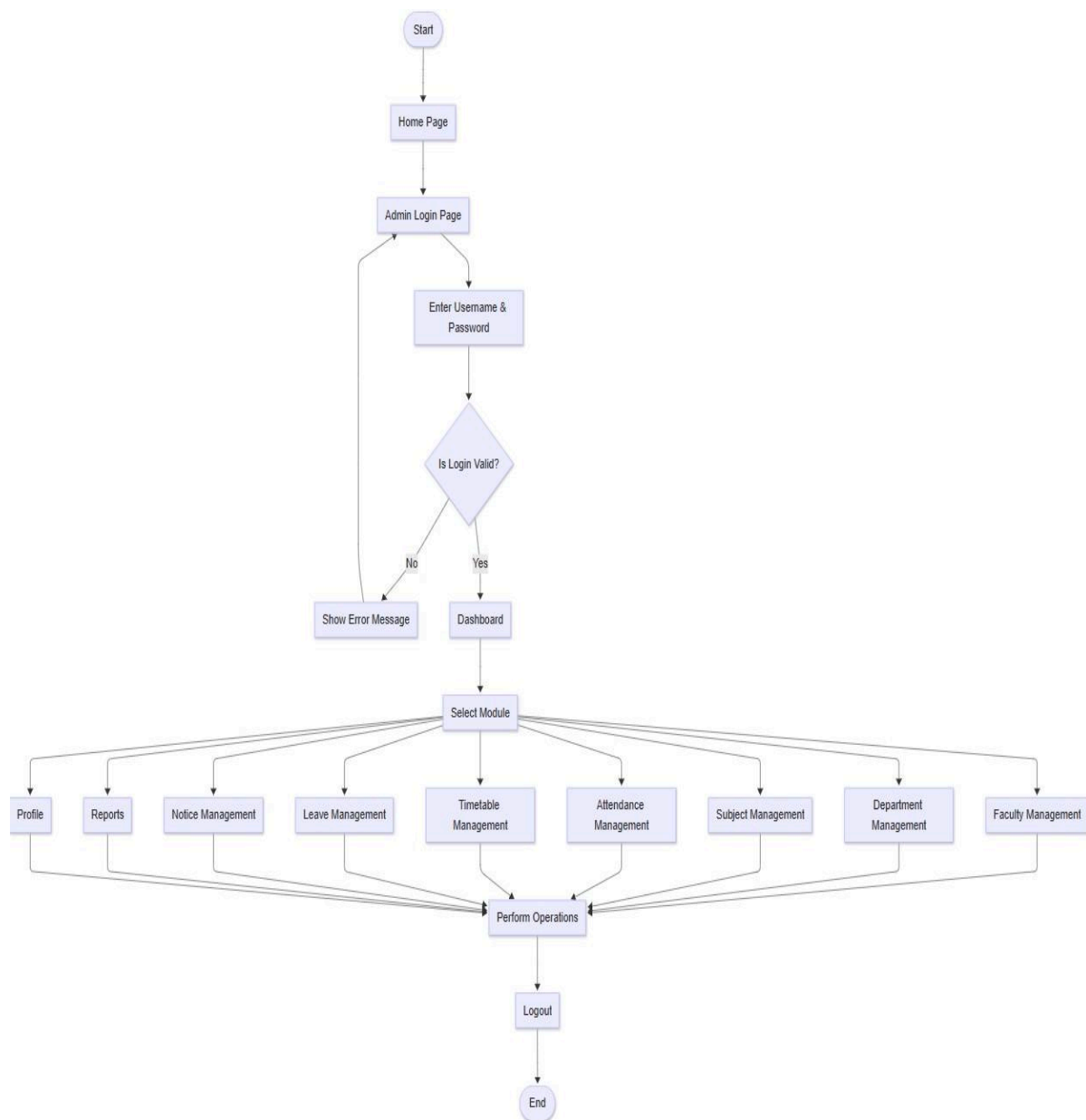


Figure 5.5: Flowchart of Faculty Management Hub System